

Abhishek Saikia

abhisheksaikia766@gmail.com — +91 70026 43704 — [GitHub](#) — [LinkedIn](#) — [Portfolio](#) — [Blogs](#)

Backend and infra-focused engineer with production Go experience, merged contributions to Athens (Go module proxy, 4.7k stars) and go-swagger (6k stars), and hands-on work building distributed systems — web crawlers, video processing pipelines, and observability tooling. Graduating 2026, targeting backend and infra roles.

Open Source Contributions

Athens — Go Module Proxy (4.7k+ stars) [PR #2093](#) Mar 2026

- Identified gaps in proxy observability; implemented 3 production metrics (cache hit rate, upstream fetch count, fetch latency) with low-cardinality labels compatible with Prometheus, Datadog, and Stackdriver in a production Go codebase.
- Metrics cover cache lookup totals, upstream fetch totals, and upstream fetch duration seconds — giving operators full visibility into proxy behavior for the first time.

go-swagger — OpenAPI Tooling (6k+ stars) [PR #3288](#) Mar 2026

- Eliminated a runtime panic in production schema parsing caused by unsafe out-of-bounds access on enum comment declarations; added guard checks preventing slice out-of-range crashes.
- Implemented a regression test reproducing the failure, ensuring the fix holds across future changes in a 6k-star Go toolchain used by thousands of developers.

Projects

Craw — Developer-Focused Web Crawler [GitHub](#) Mar 2026

- Built a production-grade web crawler in Go with distributed-system design patterns — Redis-backed crawl frontier, concurrent worker pipelines, and SimHash deduplication — achieving 3.16 pages/sec sustained over 3+ hours.
- Designed a Redis frontier with prioritized queues, retry scheduling, and visibility timeouts (SQS-like semantics) to manage crawl state and backpressure across concurrent workers.
- Implemented domain-aware rate limiting, robots.txt compliance, and failure handling using goroutine worker pools; processed 9.8GB of data with 30% duplicate filtering.
- Built dual-layer deduplication: URL normalization + hashing for exact duplicates, and SimHash with Hamming distance for near-duplicate content detection.
- Stored raw documents in MinIO (S3-compatible object storage) with structured metadata for future indexing and search ranking.

Video Transcoding Monorepo (Team project) — Scalable Video Processing Pipeline [GitHub](#) Oct 2025-Apr 2026

- Implemented the outbox pattern with concurrent-safe polling using SKIP LOCKED in PostgreSQL, ensuring reliable job delivery without dual-write failures.
- Designed a BullMQ + Redis pipeline for distributed job processing, decoupling upload ingestion from transcoding workers.
- Transcoded videos to HLS format (360p–1080p) using FFmpeg with outputs stored in S3-compatible storage.
- Containerized all services with Docker Compose for reproducible local development across a Turborepo monorepo.

Vynsol — Backend API for a Financial Dashboard [GitHub](#) 2026

- Designed and built from scratch a backend system for a financial dashboard, implementing role-based access control (RBAC), stateless JWT authentication, and middleware-driven rate limiting.
- Built offset-based pagination and background cleanup workflows using a layered architecture for security and separation of concerns.

Experience

SDE Intern — Rablo.in (Remote) Mar 2025 – Jul 2025

- Built an event-driven email automation service in Node.js handling onboarding workflows, eliminating manual steps in the activation pipeline and measurably improving activation rates.
- Designed REST API workflows and server-side state architecture, reducing client response latency by 15% through improved request handling and payload structure.
- Optimized production frontend performance via code-splitting, memoization, and SSR tuning, improving page load times by 20
- Resolved 50+ bugs across frontend and backend codebases, improving CI/CD pipeline stability and reducing build failures in production.

Education

B.Tech in Computer Science & Engineering, Jorhat Engineering College 2022 – 2026 *CGPA: 8.53/10*

Skills

Primary: Go, PostgreSQL, Redis, Docker, Linux, Git, AWS, Prometheus, BullMQ, Javascript.

Also familiar with: TypeScript, Python, Express.js, MongoDB